

Energy Conservation Law and its Application for the Direct Energy Method of Power System Stability

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Abstract: Many papers have recently contributed to the development of energy functions for power systems. However, there exists limitation in that the conventional energy functions cannot reflect the control effects of the speed governor and exciter efficiently. In this paper, two useful theorems are developed regarding the relationships of the energy integral of generator power versus its phasor current and voltage. With the use of the two theorems, the energy conservation law can be derived to reflect the resistance, flux-decaying effects, saliency with wide adaptability to various detailed generator models including the governor and exciter control loops. By the rigorous mathematical analysis and numerical tests with its application to sample systems, the proposed energy conservation law is proven to be very exact and valid for any kind of generator models including the speed governor and exciter controls.

Keywords: power system, stability, energy function, energy conservation law, exciter, governor, flux-decaying effect

I. Introduction

Most of the energy functions in earlier papers have limitations in view of the fact that the classical model of the generator is being adopted with the use of a lossless power system. With the introduction of a structure preserving energy concept [1] and center-of-mass energy analysis, considerable progress has recently been made in the development of energy functions to take into account the effects of reactive powers and transmission line resistances. One of the most recent achievements is that a well-defined energy function reflecting the line resistances is successfully developed for the multi-machine systems under an assumption of the uniform R/X ratios of the transmission lines [2,3]. Many authors have also contributed to development of the energy functions associated with detailed generator models [4,5]. On the other hand, there have been some attempts to develop an energy function reflecting the control effects of the speed governor and exciter [6,7]. However, these approaches are mainly based on the case-by-

case intuitive approach, and fail to provide with a systematic approach to develop energy functions.

In this paper, a complex integral is obtained by integrating bus current equations with respect to bus voltages. From this current-related complex integral, the energy conservation law can be derived with the use of the two new theorems.

With the rigorous mathematical analysis, it is proven that the proposed energy conservation law is very exact and valid for any kind of generator model which assumes the stator/network transients being negligible. Moreover, it should be noted that the proposed energy conservation law holds even for any generator model including the speed governor and the exciter control loops.

Finally, numerical tests have been performed to show the exactness of the energy conservation law for various cases.

II. Integral Relationships between Generator Mechanical Input and Electric Output

The energy function for power systems involves the energy stored in generators, and the system stability is mostly dominated by the generator energy. Most of earlier studies adopted the classical model of generator to derive energy function in order to avoid mathematical complexity. However, it is increasingly required to take into account the flux decaying effects, generator saliency and the effects of exciter/governor controls. In this context, integral relationships between generator mechanical input and electrical output are investigated with the detailed model of generator for the energy function derivation. This study mainly follows the detailed models proposed by Sauer [8].

For a multibus system, we have two references, system and machine, for each generator in the phasor representation. The phasor representations based on these two references have the following relationships.

$$\mathbf{E}_{Gi}' = (\mathbf{E}_{di}' + j\mathbf{E}_{qi}') e^{j(\delta_i - \pi/2)} \quad (1.a)$$

$$\mathbf{V}_{Gi} = (\mathbf{V}_{di} + j\mathbf{V}_{qi}) e^{j(\delta_i - \pi/2)} \quad (1.b)$$

$$\mathbf{I}_{Gi} = (\mathbf{I}_{di} + j\mathbf{I}_{qi}) e^{j(\delta_i - \pi/2)} \quad (1.c)$$

$$\Psi_{Gi} = (\Psi_{di} + j\Psi_{qi}) e^{j(\delta_i - \pi/2)} \quad (1.d)$$

By using the above relations, the following theorems can be developed regarding the integral relationships between the mechanical input and the electrical output in a generator. The pu(per unit) unit system is adopted throughout the mathematical development.

Theorem 1

The energy integral of generator output power can be represented by a complex line integral of current and voltage phasors as follows:

$$\oint_{V_0, \theta_0}^{(V, \theta)} (P_{Gi} d\theta_i + \frac{Q_{Gi}}{V_i} dV_i) = \int_{V_{Gi0}}^{V_{Gi}} \text{Im}(\mathbf{I}_{Gi}^* d\mathbf{V}_{Gi}) \quad (2)$$

where $\mathbf{V}_{Gi} = V_i e^{j\theta_i}$

P_{Gi}, Q_{Gi} : real and reactive generator output

power i.e. $P_{Gi} + jQ_{Gi} = \mathbf{V}_{Gi} \mathbf{I}_{Gi}^*$

Theorem 2

If the stator/network transients are negligible for Generator i , the following holds:

$$\oint_{V_0, \theta_0}^{(V, \theta)} (P_{Gi} d\theta_i + \frac{Q_{Gi}}{V_i} dV_i) = \int_{\delta_{oi}}^{\delta_i} P_{ei} d\delta_i + \int_{V_{Goi}}^{V_{Gi}} (I_{di} dV_{qi} - I_{qi} dV_{di}) \quad (3)$$

$$\text{with } P_{ei} = \frac{\omega^+}{\omega_s} (I_{qi} \psi_{di} - I_{di} \psi_{qi}) \quad (4)$$

ω_s : rated angular velocity in [pu]

$\omega^+ = \omega_s + \omega$ (Note : $P_{ei} = P_{Gi}$ if r_{si} is zero)

Proofs of the above theorems can be found in Reference [2].

Here it is noted that the assumption in the above theorem imposes few restrictions on applications of the theorem, since most of the detailed models are developed under this assumption [4,8,10].

On the other hand, we have the following generator power relation:

$$P_{ei} = P_{mi} - M_i \dot{\omega}_i - D_i \omega_i \quad (5)$$

with D_i : Generator Damping Coefficient

By using Theorem 2 and (5), one can easily get

$$\begin{aligned} & \oint_{V_0, \theta_0}^{(V, \theta)} (P_{Gi} d\theta_i + \frac{Q_{Gi}}{V_i} dV_i) \\ &= \int_{\delta_{oi}}^{\delta_i} P_{mi} d\delta_i - \frac{1}{2} M_i (\omega_i^2 - \omega_{i0}^2) - \int D_i \omega_i^2 dt \\ &+ \int_{V_{Goi}}^{V_{Gi}} (I_{di} dV_{qi} - I_{qi} dV_{di}) \end{aligned} \quad (6)$$

It is noted that the first three terms in (6) are the same as the generator-related terms in the energy function.

III. Derivation of Energy Conservation Law

For a multibus network, the bus current equation can be represented as follows:

$$\mathbf{Y}_{BUS} \mathbf{V}_{BUS} - \mathbf{I}_G + \mathbf{I}_L \equiv 0 \quad (7)$$

where $\mathbf{I}_G$: Vector of Generator Phasor Currents

(n -dimensional) with $\mathbf{I}_{Gi} = 0$ for $i > m$

$\mathbf{I}_L$: Vector of Load Phasor Currents

(n -dimensional)

All the generator buses are numbered first and m denotes the total number of generator buses. By using the above equation, an energy conservation law can be derived for power systems with the use of Theorem 1.

To begin with, we will take complex integrals of both sides of the bus current equation in (7). That is,

$$\mathbf{W} = \oint_{\mathbf{V}_{BUS0}}^{\mathbf{V}_{BUS}} [\mathbf{Y}_{BUS} \mathbf{V}_{BUS} - \mathbf{I}_G + \mathbf{I}_L]^* d\mathbf{V}_{BUS} \equiv 0 \quad (8)$$

where integral path c is given by the system trajectory.

The above equation can be rewritten in the summation form as follows:

$$\mathbf{W} = \oint_{\mathbf{V}_{BUS0}}^{\mathbf{V}_{BUS}} \left[\sum_{i=1}^N \left(\sum_{j=1}^N \mathbf{Y}_{ij}^* \mathbf{V}_j \right) d\mathbf{V}_i - \sum_{i=1}^m \mathbf{I}_{Gi}^* d\mathbf{V}_i + \sum_{i=1}^N \mathbf{I}_{Li}^* d\mathbf{V}_i \right] \equiv 0 \quad (9)$$

On the other hand, we have the following general relationships:

$$\mathbf{V}_i = V_i e^{j\theta_i} \quad (10.a)$$

$$\mathbf{I}_{Gi}^* = \frac{P_{Gi} + jQ_{Gi}}{V_i} = \frac{P_{Gi} + jQ_{Gi}}{V_i} e^{-j\theta_i} \quad (10.b)$$

$$\mathbf{I}_{Li}^* = \frac{P_{Li} + j(Q_{Li} - Q_{Gi})}{V_i} = \frac{P_{Li} + j(Q_{Li} - Q_{Gi})}{V_i} e^{-j\theta_i} \quad (10.c)$$

$$\mathbf{Y}_{BUS} = [\mathbf{G}_{ij} + j\mathbf{B}_{ij}] \quad (10.d)$$

$$d\mathbf{V}_i = e^{j\theta_i} dV_i + jV_i e^{j\theta_i} d\theta_i \quad (10.e)$$

where V_i : magnitude of bus voltage $\mathbf{V}_i$

θ_i : voltage magnitude and phasor angle of bus voltage $\mathbf{V}_i$ represented with respect to the system reference

By substitution the above equations into (9), we can derive the following equations for its imaginary part.

$$\begin{aligned} \mathbf{W}_{\text{IMAG}} = & - \left[\frac{1}{2} \sum_{i=1}^N B_{ii} V_i^2 + \frac{1}{2} \sum_{i=1}^N \sum_{j \neq i}^N B_{ij} V_i V_j \cos \theta_{ij} \right]_{(V_0, \theta_0)}^{(V, \theta)} \\ & + \sum_{i=1}^N \sum_{j=1}^N \oint_{V_0, \theta_0}^{(V, \theta)} G_{ij} (V_j \sin \theta_{ij} dV_i + V_i V_j \cos \theta_{ij} d\theta_i) \\ & - \sum_{i=1}^m \oint_{V_0, \theta_0}^{(V, \theta)} (P_{Gi} d\theta_i + \frac{Q_{Gi}}{V_i} dV_i) \\ & + \sum_{i=1}^N \oint_{V_0, \theta_0}^{(V, \theta)} (P_{Li} d\theta_i + \frac{Q_{Li} - Q_{Gi}}{V_i} dV_i) \equiv 0 \end{aligned} \quad (11)$$

The derivation procedures can be found in the reference [2]. Here it is noted that the identical zero in the above equation implies that the energy conservation law is related to this complex integral.

Energy Conservation Law

In (11), the energy terms associated with generator output powers P_{Gi} and Q_{Gi} , can be replaced by the energy terms associated with generator mechanical input P_{mi} by using (6), which yields the following energy conservation law.

$$\begin{aligned}
& \sum_{i=1}^m \frac{1}{2} M_i \omega_i^2 + \sum_{i=1}^N \left[-\frac{1}{2} B_{ii} V_i^2 - \frac{1}{2} \sum_{j \neq i}^N V_i V_j B_{ij} \cos \theta_{ij} \right] \\
& + \sum_{i=1}^N \int_{\theta_0}^{\theta} G_{ii} V_i^2 d\theta_i + \sum_{i=1}^N \sum_{j \neq i}^N G_{ij} \int_{V_0, \theta_0}^{(V, \theta)} \left(V_i V_j \cos \theta_{ij} d\theta_i \right. \\
& \left. + V_j \sin \theta_{ij} dV_i \right) \\
& - \sum_{i=1}^m \left[\int_{\delta_{0i}}^{\delta_i} P_{mi} d\delta_i + \int_{V_{g0i}}^{V_{gi}} (I_{di} dV_{qi} - I_{qi} dV_{di}) \right] \\
& + \sum_{i=1}^N \left[\int_{\theta_0}^{\theta} P_{Li} d\theta_i + \int_{V_0}^V \left(\frac{Q_{Li} - Q_{Ci}}{V_i} \right) dV_i \right] + \sum_{i=1}^m \left[\int_0^t D_i \omega_i^2 dt \right] \equiv C
\end{aligned} \quad (12)$$

where C is a constant reflecting the initial energy at the operating point (θ_0, V_0)

Energy Function

By exempting the time integral term in the above equation, we can obtain an energy function for a power system as follows:

$$\begin{aligned}
E = & \sum_{i=1}^m \frac{1}{2} M_i \omega_i^2 + \sum_{i=1}^N \left[-\frac{1}{2} B_{ii} V_i^2 - \frac{1}{2} \sum_{j \neq i}^N V_i V_j B_{ij} \cos \theta_{ij} \right] \\
& + \sum_{i=1}^N \int_{\theta_0}^{\theta} G_{ii} V_i^2 d\theta_i + \sum_{i=1}^N \sum_{j \neq i}^N G_{ij} \int_{V_0, \theta_0}^{(V, \theta)} \left(V_i V_j \cos \theta_{ij} d\theta_i \right. \\
& \left. + V_j \sin \theta_{ij} dV_i \right) \\
& - \sum_{i=1}^m \left[\int_{\delta_{0i}}^{\delta_i} P_{mi} d\delta_i + \int_{V_{g0i}}^{V_{gi}} (I_{di} dV_{qi} - I_{qi} dV_{di}) \right] \\
& + \sum_{i=1}^N \left[\int_{\theta_0}^{\theta} P_{Li} d\theta_i + \int_{V_0}^V \left(\frac{Q_{Li} - Q_{Ci}}{V_i} \right) dV_i \right]
\end{aligned} \quad (13)$$

Most of the terms in the above energy function are the same as those in the structure-preserved energy function.

Seminegativity of the Time-Derivative of Energy Function

With the use of the energy conservation law (12), the energy function (13) can be expressed in an alternative way as follows:

$$E = C - \sum_{i=1}^m \int D_i \omega_i^2 dt \quad (14)$$

with $C = E(\theta_0) = E(\omega_0, \delta_0, V_0, \theta_0)$

From this expression, we can calculate the time derivative of energy function in a simple way.

$$\frac{dE}{dt} = -\sum_{i=1}^m D_i \omega_i^2 \leq 0 \quad (15)$$

We can also check the above inequality by the conventional method using the differential chain-rule

$$\begin{aligned}
\frac{dE}{dt} = & \sum_i \left(\frac{\partial E}{\partial \omega_i} \frac{d\omega_i}{dt} + \frac{\partial E}{\partial \delta_i} \frac{d\delta_i}{dt} + \frac{\partial E}{\partial V_i} \frac{dV_i}{dt} + \frac{\partial E}{\partial \theta_i} \frac{d\theta_i}{dt} \right. \\
& \left. + \frac{\partial E}{\partial V_{di}} \frac{dV_{di}}{dt} + \frac{\partial E}{\partial V_{qi}} \frac{dV_{qi}}{dt} \right) = -\sum_i D_i \omega_i^2
\end{aligned} \quad (16)$$

(Proof is given in Appendix)

The latter is the only conventional method to evaluate the time derivative of the energy function, which usually requires very complicate procedures as shown in Appendix. Here should be noted that the proposed energy conservation law provides some intuition about the semi-negativeness of time derivative, which can be utilized as a powerful tool to develop various energy functions.

IV. Applications

The energy conservation law (12) is derived under no assumptions except the stator/network transients being negligible, and thus it can be applied to develop energy functions for most of the commonly used generator models such as E_q' -model, two-axis model and four-axis model.

Moreover, the energy conservation can provide useful information for the development of energy functions to reflect the control effects by the speed governor or the AVR (Automatic Voltage Regulator) for the exciter. This paper shows two illustrative examples.

Energy Function for the E_q' -Model

Consider the derivation of energy integral for a multimachine system with the E_q' model for all generators. In this example, the generator resistances are neglected in the generator current analysis for simplicity. Then, all the generators are represented by the following model:

$$T_{d0i}' \frac{dE_{qi}'}{dt} = -E_{qi}' - (X_{di}' - X_{di}') I_{di} + E_{fdi} \quad (17)$$

$$M_i \frac{d\omega_i}{dt} = T_{mi} - E_{qi}' I_{qi} - (X_{qi}' - X_{di}') I_{di} I_{qi} - D_i \omega_i \quad (18)$$

For the above E_q' model, the generator terminal voltages are given by

$$V_{qi} = E_{qi}' - X_{di}' I_{di} = V_i \cos \delta_{Li} \quad (19)$$

$$V_{di} = X_{qi}' I_{qi} = V_i \sin \delta_{Li}$$

where $\delta_{Li} = \delta_i - \theta_i = \text{Ang}(E_{qi}') - \text{Ang}(V_i)$: Load Angle (20)

By using the above equations, we can easily get the following equation:

$$\begin{aligned}
& \int_{V_{g0i}}^{V_{gi}} (I_{di} dV_{qi} - I_{qi} dV_{di}) \\
& = \int_{E_{q0i}'}^{E_{qi}'} I_{di} dE_{qi}' - \frac{1}{2} X_{di}' (I_{di}^2 - I_{di0}^2) - \frac{1}{2} X_{qi}' (I_{qi}^2 - I_{qi0}^2)
\end{aligned} \quad (21)$$

By using (17), it can be shown that

$$\begin{aligned}
\int_{E_{q0i}'}^{E_{qi}'} I_{di} dE_{qi}' & = \int_{E_{q0i}'}^{E_{qi}'} \frac{1}{(X_{di}' - X_{di}')} \left(-T_{d0i}' \frac{dE_{qi}'}{dt} - E_{qi}' + E_{fdi} \right) dE_{qi}' \\
& = \frac{1}{(X_{di}' - X_{di}')} \left[-T_{d0i}' \int \left(\frac{dE_{qi}'}{dt} \right)^2 dt - \frac{1}{2} (E_{qi}'^2 - E_{q0i}'^2) + \int_{E_{q0i}'}^{E_{qi}'} E_{fdi} dE_{qi}' \right]
\end{aligned} \quad (22)$$

Here, we may take the following assumptions in order to get a path-independent energy function.

- i) All line resistances are negligible, and
- ii) all governors and exciters are fixed, i.e., all P_{mi} and E_{fdi} are constant, and
- iii) real loads are constant, and reactive loads are dependent only on bus voltages.

Under these assumptions, we can get rid of all the path dependent terms. After substituting (21) and (22) into the energy conservation law (12), some manipulations yield the following energy conservation equation.

$$\begin{aligned} & \frac{1}{2} \sum_{i=1}^m M_i \omega_i^2 + \frac{1}{2} \sum_{i=1}^m \left(X'_{di} I_{di}^2 + X'_{qi} I_{qi}^2 + \frac{1}{(X'_{di} - X'_{di})} E_{qi}^2 \right) \\ & + \sum_{i=1}^N \left[-\frac{1}{2} B_{ii} V_i^2 - \frac{1}{2} \sum_{j \neq i} V_i V_j B_{ij} \cos \theta_{ij} \right] \\ & - \sum_{i=1}^m \left[P_{mi} (\delta_i - \delta_{i0}) - \frac{T_{doi}}{(X'_{di} - X'_{di})} \int \left(\frac{dE_{qi}}{dt} \right)^2 dt - \frac{E_{fdi} (E'_{qi} - E_{qi0})}{(X'_{di} - X'_{di})} \right] \\ & + \sum_{i=1}^N \left[P_{Li} (\theta_i - \theta_{i0}) + \int_{\theta_{i0}}^{\theta_i} \left(\frac{Q_{Li} - Q_{Ci}}{V_i} \right) dV_i \right] + \sum_{i=1}^m \int D_i \omega_i^2 dt = C_1 \end{aligned} \quad (23)$$

From the above energy conservation, we can easily obtain the following energy function.

$$\begin{aligned} E_1 = & \sum_{i=1}^m \frac{1}{2} M_i \omega_i^2 - \sum_{i=1}^N \left[\frac{1}{2} B_{ii} V_i^2 + \frac{1}{2} \sum_{j \neq i} V_i V_j B_{ij} \cos \theta_{ij} \right] \\ & + \frac{1}{2} \sum_{i=1}^m \left[X'_{di} I_{di}^2 + X'_{qi} I_{qi}^2 + \frac{E_{qi}^2}{(X'_{di} - X'_{di})} \right] - \sum_{i=1}^m \frac{E_{fdi} (E'_{qi} - E_{qi0})}{(X'_{di} - X'_{di})} \\ & + \sum_{i=1}^N \left[-P_{mi} (\delta_i - \delta_{i0}) + P_{Li} (\theta_i - \theta_{i0}) \right] \\ & + \sum_{i=1}^N \left[\int_{\theta_{i0}}^{\theta_i} \left(\frac{Q_{Li} (V_i) - Q_{Ci} (V_i)}{V_i} \right) dV_i \right] \end{aligned} \quad (24)$$

In the above energy function, generator currents I_{di} , I_{qi} can be eliminated by using (19) and (20). If they are eliminated, the above energy function becomes just the same as the energy function presented by Padiyar [5,6]. However, the current variables are kept for simplicity. By the virtue of (23), the seminegativity of time-derivative of the above energy function can be easily proved as follows:

$$\frac{dE_1}{dt} = - \sum_{i=1}^m \left[D_i \omega_i^2 + \frac{T_{doi}}{(X'_{di} - X'_{di})} \left(\frac{dE_{qi}}{dt} \right)^2 \right] \leq 0 \quad (25)$$

In the case where the line resistances are nonnegligible, we can easily obtain the energy function by including path-dependent integral terms associated with the line conductances in (23). Here, it should be noted that an energy function can be derived for the two-axis generator model without any difficulty.

Energy Function reflecting the Speed Governor Effects

In order to show an illustrative example to reflect the speed governor effects into the energy functions, the governor-turbine system is simplified into a 1st order transfer function as shown on Fig.1

The frequency control loop can be represented by the following block diagram with a frequency feedback loop by using the first-order reduced model for the governor-turbine system. This study assumes the integration control loop be opened for all generators.

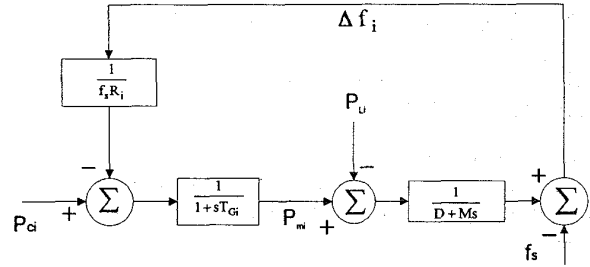


Fig. 1 First-Order Reduced Frequency Control System

From the above block diagram, we can get the following equation.

$$\Delta \dot{P}_{mi} = -\frac{1}{T_{Gi}} \Delta P_{mi} - \frac{1}{\omega_s R_i T_{Gi}} \omega_i \quad (26)$$

$$\text{with } \Delta P_{mi} = P_{mi} - P_{ci}, \quad \omega_s = 2\pi f_s, \quad \omega_i = 2\pi(f_i - f_s)$$

By using the above equation, the P_{mi} -related integral term can be evaluated as follows [7]:

$$\begin{aligned} \int_{\delta_{i0}}^{\delta_i} P_{mi} d\delta_i = & P_{ci} (\delta_i - \delta_{i0}) - \frac{1}{2} \omega_s R_i T_{Gi} (\Delta P_{mi}^2 - \Delta P_{i0}^2) \\ & - \int_{\delta_{i0}}^{\delta_i} \omega_s R_i \Delta P_{mi}^2 dt \end{aligned} \quad (27)$$

For simplicity, it will be assumed here that the position of the speed changer (P_{ci}) be fixed and Load P_{Li} constant. All bus voltages and generator internal voltages E_{qi} are also assumed to be constant. The saliency of generators are neglected by assuming $X_{qi} \equiv X'_{di}$. These assumptions have generally been accepted in the practical point of view for transient stability analysis. By substituting (27) into the energy conservation (24) under these assumptions, we can easily get the following energy function and its time derivative

$$\begin{aligned} E_2 = & -\frac{1}{2} \sum_{i=1}^m \sum_{j=1}^N B_{ij} V_i V_j \cos \theta_{ij} + \sum_{i=1}^N P_{Li} (\theta_i - \theta_{i0}) \\ & + \sum_{i=1}^m \left(\frac{1}{2} M_i \omega_i^2 - \frac{E_{qi} V_i}{X_{di}} \cos \delta_{Li} + \frac{1}{2} \omega_s R_i T_{Gi} \Delta P_{mi}^2 - P_{ci} (\delta_i - \delta_{i0}) \right) \end{aligned} \quad (28)$$

$$\frac{dE_2}{dt} = - \sum_{i=1}^m \left[\omega_s R_i \Delta P_{mi}^2 + D_i \omega_i^2 \right] \leq 0 \quad \text{for all } t \quad (29)$$

The above semi-negativity of the time derivative can be easily proven by using either of the energy conservation or

the differential chain rule.

V. Numerical Tests

In this section the energy conservation (12) is tested for the generator with various governor types and exciter types.

The energy conservation is tested by checking the energies calculated by the alternative methods given by (13) and (14). The WSCC 3-Generator system and the New England 10-Generator system are adopted as sample systems, in which the speed governor and the exciter systems for each generator are chosen by the standard type as given in the literatures [11,12]. It should be noted that the generator model includes the governor and exciter control loops.

WSCC 3-Generator System

For the contingency case, it is assumed that a three phase fault occurred near Bus #7 at $t = 0$ and was cleared by opening the line between Bus #7 and Bus #5 at $t = 0.1$ sec. The time simulation was performed for 10 sec with various time intervals.

Fig. 2 shows the results of the time simulation – the change in the generator rotor angle δ , the angular velocity ω , the mechanical output power P_m and the exciter voltage E_{fd} .

Fig. 3 shows the test results of the energy evaluation. With the normal time step 0.01 sec, the system energy has been evaluated in two different ways by (13) and (14). The two evaluated energies show slight differences (about 0.0059), which is fairly good in accuracy within tolerance 2%. Since the trapezoidal approximation method is used to integrate the path dependent terms in (13), the accuracy of numerical results depend on the integration time step. In order to get more accurate results, we have changed the time step to 0.001 sec and the results show that the two different evaluations yield the almost exactly same energy curves. This provides us with a good evidence that the proposed energy conservation law is well-satisfied.

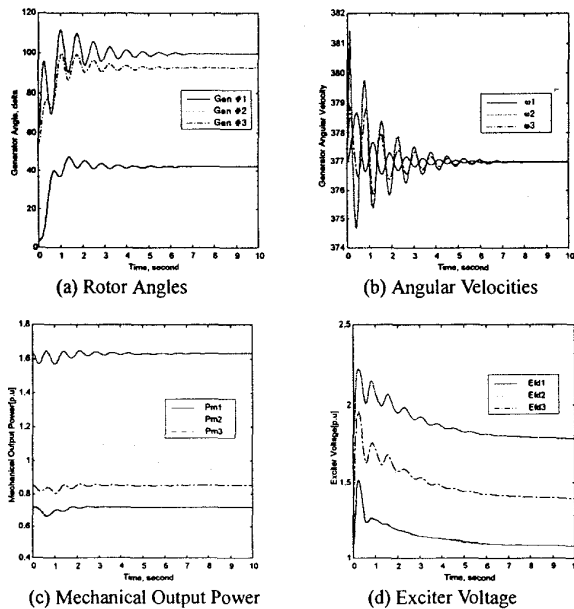


Fig. 2 Simulation Results of Generator States

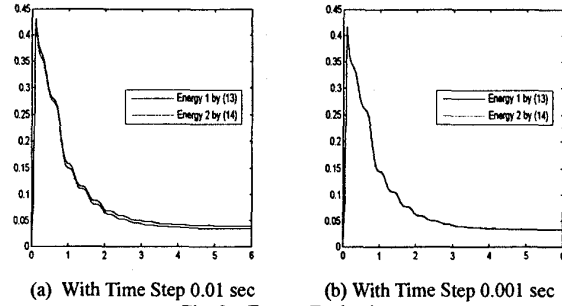


Fig. 3 Energy Evaluations

New England 10-Generator System

Various fault cases are considered with this sample system to show the validity of the proposed energy conservation law. The test simulation was always performed for 10 sec with time interval 0.001 sec.

The first contingency case is assumed that the three phase fault occurred at Bus #30 for duration of 0.1 sec. The trajectories of the generator states at Bus #30, Bus #34 and Bus #39 are shown in Fig. 4.

In this case, the system states converge rather slowly after the fault clearing, and the evaluated energies are shown in Fig. 5(a). The two evaluated energies give the almost same curves, showing that the proposed energy conservation law is well-satisfied for this sample system too.

To show the effect of the clearing time on the energy evaluations, two contingency cases are considered - energies are evaluated by changing the fault duration to 0.22 sec with the same fault bus. The results are shown in Fig. 5(b).

In the case of 0.22 sec fault duration, the energy at the clearing time has much higher than those of the cases of fault period 0.1 sec.

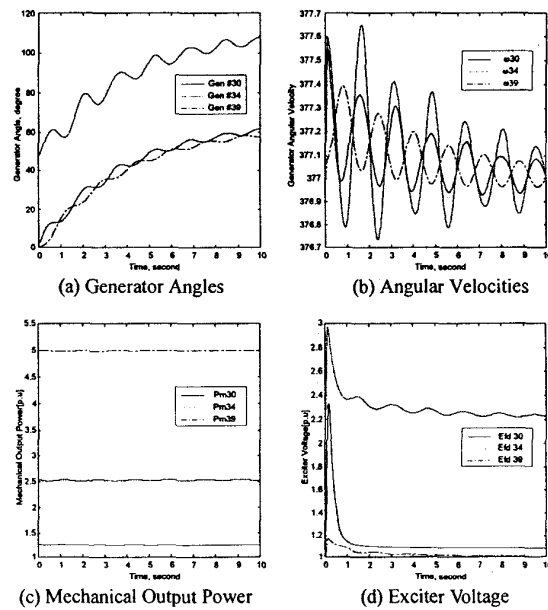
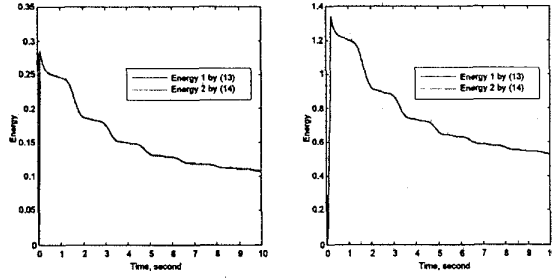


Fig. 4 Simulation Results of Generator States



(a) Fault Cleared at 0.1 sec (b) Fault cleared at 0.22 sec
Fig. 5 Energy Evaluations for Different Fault Periods

VI. Conclusions

This paper has presented a new approach to derive energy integral reflecting, flux-decaying effects, and the control effects of governor and exciter on the basis of complex line integral of the bus current equations with respect to bus phasor voltages.

In this paper, several useful theorems have been developed regarding the relationships of the energy integral of generator power versus its phasor current and voltage. A complex integral is obtained by integrating bus current equations with respect to bus voltages. From this complex-phasor integration, an exact energy conservation law is derived to reflect the governor and exciter control effects with wide adaptability to various detailed generator models.

The proposed energy conservation law has been successfully tested for the WSCC 3-Generator system and the New England 10-Generator system.

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APPENDIX

A. Proof of the Semi-Negativeness of dE/dt by (16)

We can rewrite the path-dependent integral with respect to time t . For example,

$$\int_{(\theta_0, V_0)}^{(\theta, V)} (V_i V_j \cos \theta_{ij} d\theta_i + V_j \sin \theta_{ij} dV_i) = \int_0^t \left[V_i V_j \cos \theta_{ij} \frac{d\theta_i}{dt} + V_j \sin \theta_{ij} \frac{dV_i}{dt} \right] dt \quad (A.1)$$

By using this property, the time derivative of the path-dependent integral terms can be easily obtained

By using the properties of double summation, we can obtain the following time derivative of the energy function.

$$\begin{aligned} \frac{dE}{dt} = & \sum_{i=1}^m \left(M_i \omega_i \frac{d\omega_i}{dt} - P_{mi} \frac{d\delta_i}{dt} - I_{di} \frac{dV_{qi}}{dt} + I_{qi} \frac{dV_{di}}{dt} \right) \\ & + \sum_{i=1}^N \left(\sum_{j=1}^N V_i V_j B_{ij} \sin \theta_{ij} + G_{ii} V_i^2 + \sum_{j=1}^N G_{ij} V_i V_j \cos \theta_{ij} + P_{Li} \right) \frac{d\theta_i}{dt} \\ & + \sum_{i=1}^N \left(-B_{ii} V_i + \frac{Q_{Li} - Q_{Ci}}{V_i} + \sum_{j=1}^N G_{ij} V_j \sin \theta_{ij} - \sum_{j=1}^N B_{ij} V_j \cos \theta_{ij} \right) \frac{dV_i}{dt} \end{aligned} \quad (A.2)$$

On the other hand, differentiating (6) with respect to time gives the following equation.

$$\begin{aligned} P_{Gi} \frac{d\theta_i}{dt} + \frac{Q_{Gi}}{V_i} \frac{dV_i}{dt} \\ = P_{mi} \frac{d\delta_i}{dt} - M_i \omega_i \frac{d\omega_i}{dt} - D_i \omega_i^2 + I_{di} \frac{dV_{qi}}{dt} - I_{qi} \frac{dV_{di}}{dt} \end{aligned} \quad (A.3)$$

Substituting the above equation and rearranging it yield

$$\begin{aligned} \frac{dE}{dt} = & - \sum_{i=1}^m D_i \omega_i^2 \\ & + \sum_{i=1}^N [G_{ii} V_i^2 + \sum_{j=1}^N V_i V_j (B_{ij} \sin \theta_{ij} + G_{ij} \cos \theta_{ij}) - P_{Gi} + P_{Li}] \frac{d\theta_i}{dt} \\ & + \sum_{i=1}^N [-B_{ii} V_i^2 + \sum_{j=1}^N [V_i V_j (G_{ij} \sin \theta_{ij} - B_{ij} \cos \theta_{ij}) \\ & - Q_{Gi} + (Q_{Li} - Q_{Ci})] \frac{1}{V_i} \frac{dV_i}{dt} \end{aligned} \quad (A.4)$$

with $P_{Gi} = Q_{Gi} = 0$ if $i > m$

In the above equation, the quantities in the two brackets become zeros respectively since they represent real and reactive power balance equations at each bus. Therefore, the seminegativeness of dE/dt can be proven as follows:

$$\frac{dE}{dt} = - \sum_{i=1}^m D_i \omega_i^2 \leq 0 \quad \text{for all time } t$$